

FOR REVIEW · PENDING DECISIONS

LabSolution LIS — Stage 1 & 2 Slice Decision Dossier

Human-in-the-loop review package — slices blocked on a decision or action

Branch `docs/stage-1-2-decision-dossier` · PR #25 · drafted 2026-06-29

Source of truth: `docs/decisions/stage-1-2-slice-decision-dossier.md`

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1. 0 — Read first: the SD-0 scope ruling (ADR-0008 ↔ LIS-74)
2. 1 — Decision summary table (SD-0 ... SD-9)
3. 2 — Stage-1 decisions (SD-1/3/6/8/9a)
4. 3 — Stage-2 decisions (SD-2/4/5/7/9b)
5. 4 — Suggested decision order
6. 5 — Context: non-decision Stage 1–2 slices

Stage 1 & 2 — Slice Decision Dossier (HITL)

Purpose. Consolidates the **stage-1 and stage-2 slices that are blocked on a human decision or human action** — i.e. the slices currently parked in Plane states **ready-for-human** (*requires human implementation/action*) or **needs-info** (*waiting on information*), per [docs/agents/triage-labels.md](#) . Each entry states the decision, why it matters, the options, a recommendation, the owner, and what it gates — so the call can be taken in one pass.

Scope. Stages 1–2 only. The *fully-specified agent work* (**ready-for-agent**) and the *Done / In-Progress* slices are listed in §5 for context but need no decision.

Status: drafted by an agent **2026-06-29**, **pending human review** (M. Uy = system/technical owner; A. L. Pinote = QA/regulatory owner, per [ADR-0007](#)).

Sources: [LIS_IMPLEMENTATION_PLAN.md](#) §3 (Stage 1–2), [docs/testing/stage-1-3-machine-access-checklist.md](#) (LIS-74 re-scope), [ADR-0008](#) (engine + v1 fleet), [docs/compliance/decisions-register.md](#) (DEC-04/06). Slice bodies in Plane are empty — the **title is the contract**; the detail lives in the sources above.

0. Read this first — one ruling unblocks half the list

There is a **cross-document conflict** that has to be resolved before the per-slice calls mean anything:

- ADR-0008 / DEC-06 (2026-06-27)** pins the **v1 pilot fleet** as **EDAN H60S/H99S + RAYTO RT-7600 — HL7-v2/MLLP, result-ingestion only**, and explicitly **defers ERBA EC90 (serial group), HETO AU120, and the bidirectional host-query / order-download path to post-pilot (v1.1) under change control**.
- The availability re-scope (2026-06-26, LIS-74)** — the machine-access checklist and the implementation-plan stage table — makes **ERBA EC90 the Stage-2 in-pilot vehicle** and treats the whole ASTM/serial stack as pilot work.

These disagree. Taken literally, ADR-0008 puts *all of Stage 2* (LIS-23...30) and Stage-1's bidirectional slice (LIS-18) **outside** the M1 pilot — yet they are being actively built (LIS-23 Done; LIS-24/26 urgent). **Decision SD-0 below is the meta-call** that determines whether the rest of Stage 2 is pilot-critical or post-pilot, and therefore how urgent every Stage-2 row in this dossier is.

1. Decision summary

Urgency: ● on the pilot critical path / blocks a milestone · ● needed before its stage gate · ● cheap confirmation / cleanup. **Type:** *Decide* = a real choice · *Act* = a human action (bench/procurement) with the call already implied.

ID	Slice	Decision / action	Urg	Type	Recommendation (1-liner)
SD-0	scope (ADR-0008 ↔ LIS-74)	Is Stage 2 / ERBA EC90 / bidirectional in the M1 pilot or post-pilot v1.1 ?	●	Decide	Build the ASTM stack now, keep EC90 bench-validated but post-pilot for go-live — reconcile the two docs explicitly.
SD-1	LIS-12 (S1.0)	Edge-driver transport substrate ADR (OIE vs bespoke)	●	Decide	Superseded by ADR-0008 (reuse analyzer-bridge). Rescope LIS-12 to "record the substrate ADR: how serial/file transports attach to the HTTP-fronted bridge," or close it.
SD-2	LIS-27 (S2.5)	Bidirectional ASTM coverage — source a DiaSys RESPONS or stay simulator-only?	●	Decide	Simulator-only for the pilot; only buy a bidirectional ASTM unit if a pilot site actually has one.
SD-3	LIS-19 (S1.7)	Mindray labXpert file-mode ingest — defer or wontfix?	●	Decide	Defer (keep the contract; no available file-drop unit). Clear needs-info .
SD-4	LIS-29 (S2.7)	HORRON EA2000 re-verification gate	●	Decide	Defer + reclassify — HORRON is HL7/TCP, not ASTM; it doesn't belong in Stage 2.
SD-5	LIS-24 (S2.2)	ASTM E1394 parser thread — why ready-for-human ?	●	Decide	Re-triage to ready-for-agent : build against a spec-synthesized EC90 fixture now; back-fill the real capture from the bench (SD-7).
SD-6	LIS-20 (S1.8)	EDAN H60S signed bench-conformance	●	Act	Schedule the bench session — retrieve H60S from warehouse; gates the Stage-1 "supported" milestone.
SD-7	LIS-30 (S2.8)	ERBA EC90 signed bench-conformance	●	Act	Retrieve EC90; capture the undocumented RS-232 baud/pinout; sign. (Urgency follows SD-0.)
SD-8	LIS-21 + LIS-77 (S1.9/S1.10)	HETO AU120 second-vendor confirm + bench	●	Act	On arrival (–now): confirm HL7/MLLP screen, pin port + listener role, resolve 2.3.1-vs-2.5, then bench.
SD-9	LIS-78 + LIS-79 (S1.11/S2.9)	EDAN H99S + Seamaty SD1 missing specs	●/●	Act	H99S: read nameplate first, then chase EDAN — it's a v1 anchor (DEC-06), so it's on the pilot path. SD1: request the spec, keep provisional .

2. Stage-1 decisions

SD-1 · LIS-12 [S1.0] — Edge-driver transport substrate ADR · ready-for-human, high

Decision. Choose the transport substrate ("OIE channels vs bespoke drivers") and record an ADR.

Status / why this is mostly already decided. [ADR-0008](#) (DEC-04) already settled the engine: **reuse** openelis-analyzer-bridge — *not* a Mirth/OIE fork, *not* bespoke-from-scratch. So LIS-12 as titled is **superseded**. What is **genuinely still open** is narrower and worth an ADR: the bridge is the **"ASTM-HTTP Bridge"** — it is **HTTP-fronted**, so the substrate question is *how the non-HTTP transports attach to it*:

- **HL7/MLLP (Stage 1):** does the bridge listen MLLP directly, or do we front it with a thin MLLP → HTTP shim?
- **Serial/ASTM (Stage 2) and the SnibeLis PC (Stage 3):** what feeds the bridge — a serial-listener sidecar, a file/DB poller, or a per-channel adapter?

Options. (a) **Rescope LIS-12** to "record the substrate ADR for how MLLP / serial / file transports attach to the analyzer-bridge" (recommended); (b) close LIS-12 as resolved-by-ADR-0008 and fold the substrate note into each stage's edge slice; (c) re-open the engine choice (not recommended — ADR-0008 is Accepted).

Recommendation. (a) — keep the slice but rescope its title; the channel/transport attachment is a real, undecided design point that the conformance + threat surface (REQ-SEC-03 channel isolation) depends on.

Owner. M. Uy (system owner) · **Gates.** the validated edge boundary, REQ-SEC-03 wording, Stage-2/3 edge slices.

SD-3 · LIS-19 [S1.7] — Mindray labXpert file-mode (shared-folder) ingest · needs-info, medium

Decision. Keep, defer, or drop the file-drop ingestion path.

Context. "labXpert" is **Mindray middleware, not an analyzer**, and **no available unit speaks file-mode** — EDAN H60S and HETO AU120 are MLLP-only. So nothing in the current fleet can exercise this path.

Options. (a) **Defer** — keep the slice + the file-mode contract for when a file-drop unit appears (e.g. MEDICA EasyStat's proprietary text dump); (b) **wontfix**; (c) build it now against a synthetic file-drop fixture.

Recommendation. (a) **Defer, not wontfix** — file-drop is a real ingestion pattern we'll meet again. Move out of needs-info with the note "no available file-mode unit; parked under change control (REQ-QMS-03)."

Owner. M. Uy · **Gates.** nothing on the pilot path.

SD-6 · LIS-20 [S1.8] — EDAN H60S signed bench-conformance · ready-for-human, high ●

Action (call is implied). Run the physical-instrument bench-conformance and sign the report.

Context. This is the **gate from "code-green" to "supported"** for the Stage-1 anchor analyzer, and the real-world half of the Stage-1 *first-result* milestone (the E2E LIS-17 is In Progress). H60S is **in the warehouse** — retrieve it, stand up the edge listener on TCP 7999, confirm direction + a sample result + raw archive, and sign.

Decision content. Who runs it and when; firmware ≥ APP V1.10; cable/switch logistics.

Owner. M. Uy (validation lead) + bench operator · **Gates.** Stage-1 "supported" matrix row; the milestone demo.

SD-8 · LIS-21 [S1.9] + LIS-77 [S1.10] — HETO AU120 second-vendor · ready-for-human ●

Action. On arrival, confirm the interface (LIS-77), then run + sign bench-conformance (LIS-21).

Context. AU120 was **"incoming ~next week" as of 2026-06-26** → likely arriving the week of this dossier. It is the **second-vendor HL7/MLLP** unit that proves the parser isn't EDAN-specific. Open unknowns to resolve **on arrival**: (i) does the AU120 expose the same **HETO AU-series HL7/MLLP** screen (KB only documents the AU400 / Konig AP300)? (ii) **pin the TCP port + listener role** (the manual omits both); (iii) resolve the **HL7 2.3.1-vs-2.5** wire version.

Decision content. Confirm arrival; assign the on-arrival capture. *Note the SD-0 scope question:* under a strict ADR-0008 reading AU120 is a **deferred** unit — confirm whether the second-vendor proof is in the pilot or post-pilot.

Owner. M. Uy + bench operator · **Gates.** the "vendor-tolerance" exit-gate item for Stage 1.

SD-9a · LIS-78 [S1.11] — EDAN H99S nameplate + LIS spec · ready-for-human ●

Action. Read the unit's **nameplate / serial** to disambiguate the model (it is **not in the KB**, and "H99S" is not a recognized EDAN catalog model — documented EDAN siblings are H60S/I15/M16), then obtain the LIS interface spec from EDAN.

Why it matters. DEC-06 pins H99S as a **v1 pilot anchor** (EDAN H60S/H99S + RT-7600), so a missing protocol spec is **on the pilot critical path**, not a nice-to-have. The cheap first move (read the nameplate) may collapse this to "it's an H60S/M16, we already have the doc."

Recommendation. Read the nameplate **now** (free); only open a vendor-spec request to EDAN if it's a genuinely distinct model.

Owner. M. Uy + LabSolution warehouse · **Gates.** the H99S driver confirm (one of ADR-0008's four **[NEEDS-HUMAN]** protocol confirms).

3. Stage-2 decisions

All Stage-2 urgencies are conditional on SD-0. If Stage 2 is post-pilot (strict ADR-0008), drop these to 🟡 and sequence after the pilot.

SD-2 · LIS-27 [S2.5] — DiaSys bidirectional Q-record thread · **needs-info** , medium

Decision. Source a **bidirectional ASTM analyzer** (e.g. DiaSys RESPONS 920) for the live host-query / NAK-retransmit negatives, or keep that path **simulator-only**?

Context. ERBA EC90 is **upload-only** — it cannot exercise the Q-record query or the NAK-retransmit path on the bench. The richest bidirectional ASTM spec (DiaSys 920) is **not in the available fleet**. The ASTM simulator harness (LIS-25) can drive both paths in CI without hardware.

Options. (a) **Simulator-only** — keep the bidirectional/NAK path proven against LIS-25, mark LIS-27 simulator-scoped, defer the live bidirectional test under change control (matches the plan's stated stance); (b) **acquire a DiaSys RESPONS** (cost + lead time) for a real bench bidirectional test.

Recommendation. (a) for the pilot — ADR-0008 already defers bidirectional/order-download to v1.1, so a simulator-validated path is sufficient for M1. Only buy a bidirectional unit if a **pilot site actually operates one**.

Owner. M. Uy + Pinote (procurement) · **Gates.** the Stage-2 bidirectional exit-gate item (currently simulator-driven).

SD-4 · LIS-29 [S2.7] — HORRON EA2000 re-verification gate · **needs-info** , medium

Decision. Defer, and **reclassify out of Stage 2**.

Context. HORRON EA2000 is **not available**, and the cross-check found it is **HL7/TCP native (or legacy serial+middleware)**, **not ASTM** — so even when acquired it belongs in the **Stage-1 (HL7)** group, not Stage 2. Its image-native source PDF carries a **re-verification gate** (driver not merged until the source PDF is re-verified and signed).

Recommendation. **Defer**; record "not available; re-files under Stage 1 (HL7) when acquired; source-PDF re-verification gate retained." Clears **needs-info**.

Owner. M. Uy · **Gates.** nothing on the pilot path.

SD-5 · LIS-24 [S2.2] — ASTM E1394 parser thread · **ready-for-human** , urgent

Decision. Confirm whether this genuinely needs a human, or can proceed as agent work.

Context. A typed-record-tree ASTM E1394 parser ($H \rightarrow P \rightarrow O \rightarrow R \rightarrow L$, tolerant of spec deviation) is **agent-implementable given a fixture**. The likely reason it sits in **ready-for-human** is that it wants a **real captured ERBA EC90 frame** as the fixture — which depends on the bench (SD-7). But the low-level codec (LIS-23) is already **Done**.

Options. (a) **Re-triage to ready-for-agent** — build the parser against a **spec-synthesized** EC90 fixture now (it's marked *urgent*), and back-fill the real captured frame from the SD-7 bench session; (b) hold it for the physical capture (serializes the urgent work behind hardware retrieval).

Recommendation. (a) — don't block urgent parser work on hardware; validate against the spec-derived fixture, then add the captured-frame fixture as a conformance back-fill.

Owner. M. Uy · **Gates.** the Stage-2 parser deliverable (LIS-26 channel thread depends on it).

SD-7 · LIS-30 [S2.8] — ERBA EC90 signed bench-conformance · **ready-for-human** , high

Action. Retrieve EC90 from the warehouse, **capture the undocumented RS-232 baud/pinout** on the bench, run conformance (direction + sample + raw), sign; record traceability rows.

Decision content. **RS-232 vs Ethernet** path (both supported by the unit); who runs it and when. Urgency inherits from **SD-0** (pilot vs post-pilot).

Owner. M. Uy + bench operator · **Gates.** the Stage-2 "supported" matrix row.

SD-9b · LIS-79 [S2.9] — Seamaty SD1 LIS spec · ready-for-human

Action. Request an **SD1-specific** LIS/communication spec from Seamaty (the KB has only the **SG1**, a *different* instrument — handheld blood-gas/ISE); meanwhile photograph the SD1's LIS screen + capture a frame.

Context. Unlike H99S, **SD1 is *not* a pinned v1 anchor** — DEC-06 lists it as "pending protocol / confirm before placing." So it stays **provisional** (Stage-1 *iff* it turns out HL7) and must **not block the pilot**.

Recommendation. Open the vendor request for lead-time, keep SD1 out of the pilot critical path until its protocol is known.

Owner. Pinote (vendor liaison) + M. Uy · **Gates.** nothing on the pilot path (provisional unit).

4. Suggested decision order

- 1. **SD-0** (scope) — one call that re-prioritizes every Stage-2 row. *Decide first.*
- 2. **SD-6 / SD-9a** — H60S bench + H99S nameplate: both on the **Stage-1 pilot critical path**; H60S gates the milestone, H99S is a v1 anchor with no spec.
- 3. **SD-8** — HETO AU120 on-arrival confirm (time-boxed by the delivery, ~now).
- 4. **SD-1, SD-5** — cheap re-triage/rescope calls that unblock active work (transport ADR; LIS-24 to agent).
- 5. **SD-2, SD-3, SD-4, SD-7, SD-9b** — defer/confirm + procurement-lead-time items; resolve after SD-0 sets their urgency.

5. Context — non-decision Stage 1–2 slices (no action needed)

Slice	State	Note
LIS-13 / S1.1 — MLLP frame + ACK^R01	Done	transport merged
LIS-14 / S1.2 — H60S ORU^R01 → LOINC/UCUM	Done	
LIS-15 / S1.3 — append-only Result store	Done	
LIS-16 / S1.4 — raw-archive + replay round-trip	Done	
LIS-17 / S1.5 — milestone E2E (H60S → Result + ACK)	In Progress (urgent)	the Stage-1 <i>first-result</i> milestone, in flight
LIS-18 / S1.6 — H60S bidirectional QRD/QRF	ready-for-agent	⚠ ADR-0008 defers bidirectional to v1.1 — see SD-0
LIS-23 / S2.1 — ASTM E1381 codec	Done	low-level framing/checksum
LIS-25 / S2.3 — ASTM simulator harness	ready-for-agent	drives the bidirectional/NAK negatives for SD-2
LIS-26 / S2.4 — ERBA EC90 channel thread	ready-for-agent	depends on LIS-24 (SD-5)
LIS-28 / S2.6 — EC90 branched channel → electrolyte Result	ready-for-agent	

Cross-references: [LIS_IMPLEMENTATION_PLAN.md §3](#), [docs/testing/stage-1-3-machine-access-checklist.md](#), [ADR-0008](#), [docs/compliance/decisions-register.md](#). RT-7600 wire-format capture (LIS-76, a v1 anchor with unconfirmed format) is Stage 3 — tracked there, but note it is also on the pilot fleet path.